

Question	Answer	Marks	Guidance
3(a)	(when two or more) waves meet / overlap (at a point)	B1	Allow 'waves intersect / merge' Ignore 'waves superpose / interfere / combine / interact' Allow answer given in the context of two waves forming a stationary wave. E.g. 'waves travelling in opposite directions'
	(resultant) displacement is sum of the individual displacements	B1	Allow 'resultant wave' or 'total displacement' for 'resultant displacement' Ignore just 'resultant displacement is the sum of the waves' Ignore 'sum of displacement is the sum of the individual displacements' but <u>allow</u> 'sum of resultant displacement is the sum of the individual displacements' Not 'amplitude' instead of 'displacement' ignore 'waves of same speed / frequency / wavelength' ignore the word 'algebraic' (as in 'algebraic sum') ignore 'coherent / identical waves'
3(b)(i)	constant phase difference	B1	Ignore 'same frequency/wavelength/speed' Ignore 'have the same phase difference' Ignore 'phase relationship' for 'phase difference' Ignore 'in phase'
3(b)(ii)	(the light) diffracts	B1	Ignore 'spreads'

Question	Answer	Marks	Guidance
3(b)(iii)	waves (from each slit) meet / superpose (at the screen)	B1	Allow 'overlap / superimpose / interfere / combine' for 'superpose' Ignore 'interact / collide / coincide / intersect'
	waves have 0 path difference / 0 phase difference	B1	Allow 'in phase' Ignore 360°
	constructive interference occurs / (resultant wave has) maximum amplitude	B1	Ignore 'intensity is maximum' Ignore references to nodes and antinodes
3(b)(iv)	$\lambda = ax / D$	C1	
	$x = (560 \times 10^{-9} \times 0.43) / (1.4 \times 10^{-5})$ = 0.017	C1	
	distance = 0.017 / 2 = 0.0086 m	A1	
3(c)(i)	<i>either:</i> B marked above Y at same distance above Y as XY <i>or:</i> B marked below 0.43 m arrow, at a distance from X that is 2XY	B1	tolerance ± 3 mm
3(c)(ii)	D marked below X at distance below X equal to XY	B1	tolerance ± 3 mm
3(d)	$c = f\lambda$	C1	Allow v for c
	frequency = $(3.00 \times 10^8) / (560 \times 10^{-9})$ = 5.4×10^{14} Hz	A1	ECF POT error in the wavelength from 3(b)(iv) if same POT error made